Activity-1 : The "school cafteria" menu survey

Phase: Data collection & Preparation

objectives: learn to design unbiased questions, collect raw data, and clean it for analysis

1. Design the Survey Instrument

Guidelines for unbiased questions:

- Use clear and simple Language
- Provide equal options
- Allow one clear response

Section A: Basic Details

1. Grade:
2. Gender (optional):

Section B: Menu Preference

1. Which new dish would you most likely try?
   A) Veggie wrap
   B) chicken Burger
   C) Pasta Salad
   D) None of these

2. How often do you eat in the cafeteria?
   - Daily
   - 3-4 times/week
   - Rarely

3. What matters most to you?
   - Taste
   - Price
   - Healthiness
   - Portion size

2. Data collection

- Survey 30-50 students from different grades
- Ensure
  - Equal representation
  - Honest responses
  - No Influencing answers
- Record responses immediately to avoid mistakes

3. Data Entry in Spreadsheet

use:-

- microsoft Excel
- Google sheets

Example Data table:-

Student ID	Grade	Dish selected	frequency	priority
1	8	chicken burger	Daily	Taste
2	7	veggie wrap	Rarely	Healthiness
3	9	chicken Burger	3-4 times week	Taste

4. Cleaning the data

Common Issues

1. Spelling mistakes:-
   "Chicken Burger" → correct to "chicken Burger".
2. Different formats
   "veggie wrap" → Standardize to "veggie wrap"

distinct Department_ID from the employees file.

Algorithm:

3. Extra Spaces
"Pasta Salad" → Remove Spaces
4. Blank entries
· Fill if known or mark as missing
5. Duplicate entries
· Remove repeated student responses

5. Standardization Rules
· use consistent capitilization (e.g. Title case)
· keep options exactly as listed in survey
· use data validation drop down in excel sheet to prevent mistakes.

6. Final outcome:
After cleaning:

Student ID	Grade	Dish selected	Frequency	Priority
1.	8	chicken burger	Daily	Taste
2.	7	veggie wrap	rarely	Healthiness
3.	9	chicken burger	3-4 times/week	Taste

Data is now:
· organized
· Consistent
· Error-free
· Ready for Analysis.